

COURSE SYLLABUS

CS300 – Elements of Software Engineering

1. GENERAL INFORMATION

Course name:	Elements of Software Engineering
Course name (in Vietnamese):	Nhập môn công nghệ phần mềm
Course ID:	CS300
Knowledge block:	Fundamentals
Number of credits:	4
Credit hours for theory:	40
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	
Prior-course:	
Instructors:	Nguyễn Thị Minh Tuyền

2. COURSE DESCRIPTION

The course is designed to provide students the concepts, principles, and practices of software engineering, focusing on software requirements specification, analysis & design, implementation, testing, integration, maintenance, and management. The students will also have opportunities to apply software engineering principles, methods, techniques, and tools by working on multi-person teams to develop and deliver software products.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Work individually and in group to present reports based on available templates	
G2	Explain software concepts in English.	
G3	Explain basic concepts, principles, methods, and techniques in software engineering	
G4	Explain and classify different software processes	
G5	Set up each phase while developing software	
G6	Construct software artifacts in each phases in software process.	
G7	Use CASE tools	

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Set up, organize operate and manage team	IT
G1.2	Participate in group discussion on course topics	U
G1.3	Analyze, synthesize and write technical reports based on available templates	IT
G2.1	Explain course concepts in English.	I
G2.2	Summarize course session in English.	I
G3.1	Explain basic concepts in Software engineering: software, system, process, main activities in a SDCL, SRS, diagrams.	IT

G3.2	Recognize roles, describe responsibilities and professional ethics in Software engineering	I
G3.3	Improve skills to update new knowledge, self-study, self-development and adaptation	I
G3.4	Establish career skills	I
G4.1	Differentiate software processes: waterfall model, incremental development, reuse-oriented software engineering	IT
G5.1	Describe main activities in a process: requirements engineering, design, implementation, V&V, evolution	IT
G5.2	Outline software process model in analysis & design activities	ITU
G5.3	Explain the importance of project management	I
G5.4	Practice software phases in SDLC	ITU
G6.1	Construct software artifacts for a small and medium-sized software in a systematic way.	TU
G7.1	Use CASE tools in developing software (Slack, Trello, GitHub, Visual Paradigm, IDEs, etc.)	ITU

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Course introduction Overview Watch videos #1, #2	G1.2, G2.1, G2.2, G3.1, G3.2, G3.3, G3.4, G5.4	Lecturing Q&A, Group discussion Role play (Chapter 1)
2	Software processes Watch videos #3, #4	G1.2, G2.1, G2.2, G4.1, G5.1	Lecturing Group discussion (Chapter 2)

3	Project management Watch video #6	G2.1, G2.2, G3.1, G3.4, G5.3	Role play Group Discussion (Chapters 22, 23)
4	Requirements engineering	G1.1, G1.2, G1.3, G2.1, G2.2, G3.2, G5.1, G6.1	Lecturing Role play Group discussion (Chapter 4)
5	System modeling	G1.2, G2.1, G2.2, G5.2, G6.1, G7.1	Lecturing Group discussion & presentation Exercise (Chapter 5)
6	Architectural design	G2.1, G2.2, G5.1, G5.2, G6.1, G7.1	Lecturing Q&A (Chapter 6)
7	Design and Implementation	G1.2, G2.1, G2.2, G5.2, G6.1, G7.1	Lecturing Group discussion Exercise (Chapter 7)
8	User interface design	G1.2, G1.3, G2.1, G2.2, G5.2, G6.1, G7.1	Lecturing Group discussion & design UI (Chapter 16, SE 8 th edition)
9	Software testing	G1.3, G2.1, G2.2, G5.1, G5.2, G6.1, G7.1	Lecturing Exercise (Chapter 8)

10	Agile software development Watch video #5	G2.1, G2.2, G3.1, G4.1	Case study, discussion Exercise (Chapter 3)
10	Review	G1.2, G3.1, G3.2, G3.4, G4.1, G5.1, G5.2, G5.3, G5.4, G7.1	Lecturing Q&A, Discussion Exercise

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the corresponding topic and ask students to do their lab exercises either on computer in the lab or at home. All the lab work submitted will be graded. There would be a final exam for lab work.

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			
A11	In-class participation & exercises	Active participation in group discussion, questions and answers Guided exercises for each topic learnt	G4.1, G5.1, G5.2, G5.3	10%
A12	Quizzes	Two quizzes check the knowledge comprehension	G2.1, G3.1, G3.2, G4.1, G5.1, G5.3	15%
A2	Projects			40%

A21	Project assignments PA1 – PA8	5 Students/group. Each team is required to work on a software project and submit results regularly in form of assignments. At the end of the class, each team will give an oral presentation discussing their completed work and demonstrating the software.	G1.1, G1.3, G3.2, 5.2, 5.4, 6.1, 7.1	40%
A3	Exams			35%
A31	Final exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems	G2.1, G3.1, G3.2, G4.1, G5.1, G5.2, G5.3, G5.4	35%

7. RESOURCES

Textbooks

- **Software Engineering, 10th Edition** (8th and 9th Edition is also fine), Ian Sommerville , Pearson, 2015, ISBN-13: 978-0133943030

Others

- **Software Engineering: A Practitioner's Approach, 8th Edition** , Roger S. Pressman, McGraw-Hill Higher Education, 2014, ISBN-13: 978-0078022128
- **Software Engineering with UML** , Bhuvan Unhelkar, CRC Press, 2018, ISBN-13: 978-1138297432
- IDEs
- Slack, Trello, GitHub, etc.
- Visual Paradigm
- etc.

List of videos

No.	Video name	URL
1	10 Questions to Introduce Software Engineering	https://www.youtube.com/watch?v=gi5kxGslkNc
2	Why Software Engineering Matters	https://www.youtube.com/watch?v=R3NzTt0BTWE
3	Plan-driven and Agile Software Processes	https://www.youtube.com/watch?v=q8X2Rk5sRFI
4	Fundamental Activities in Software Engineering	https://www.youtube.com/watch?v=Z2no7DxDWRI
5	Intro to Scrum in Under 10 Minutes	https://www.youtube.com/watch?v=XU0lIRltyFM
6	Software Project Management - Why it's Different	https://www.youtube.com/watch?v=TYBVAvWkG6M

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.